

Senthil K

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Skills

Languages: JavaScript, TypeScript, Go, Python, Rust

Frontend: React, Next.js, Vue.js, Nuxt.js, React Native, PWA, Three.js, GSAP, Tailwind CSS, Material UI, SCSS

Backend: Node.js, Express.js, Go Fiber, NestJS, Fastify, FastAPI

Databases & Search: PostgreSQL, Redis, MongoDB, DynamoDB, ElasticSearch

Messaging & Observability: Kafka, Grafana, Prometheus

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, Lambda, SES)

Tools & Other: WebAssembly with Rust, REST APIs, Monorepo Architecture, Git, Figma, Blender, Framer

Summary

Software Engineer with 1.8 years of experience building scalable web applications and enterprise software across frontend and backend systems. Experienced in developing high-performance React and Next.js applications, architecting secure APIs with Node.js and AWS serverless services, and optimizing large-scale data-intensive interfaces handling thousands of records. Skilled in TypeScript, PostgreSQL, DynamoDB, Docker, and modern cloud-native development, with a strong focus on performance, scalability, maintainable architecture, and delivering production-ready software for enterprise clients.

Work Experience

Software Engineer

OCT 2024 - JUNE 2026

Mako it lab, Chennai

- Delivered **16+ production applications and enterprise platform extensions** for clients across project management, geospatial analytics, healthcare, recruitment, and workflow automation, contributing throughout the software development lifecycle from requirements to production deployment.
- Designed and developed scalable React and Next.js applications handling **large-scale datasets (6,000+ records)** with virtualization, advanced filtering, pagination, reusable component architecture, and optimized rendering strategies, improving UI responsiveness by **60–70%**.
- Engineered high-performance editable data grids supporting **1,000+ rows and 15+ columns**, leveraging event delegation, state management, and rendering optimizations to reduce DOM interactions by approximately **90%**.
- Developed secure backend services and middleware using **Node.js, AWS Lambda, API Gateway, and DynamoDB**, implementing authentication, authorization, centralized business logic, and serverless integration patterns.
- Built production-ready caching strategies including scheduled refresh, on-demand invalidation, and backend caching layers to optimize application performance while reducing infrastructure costs.
- Architected complex client-side state management using **Zustand** and **TanStack Query**, minimizing unnecessary re-renders and improving interaction latency by approximately **50%**.
- Developed enterprise geospatial interfaces using **Mapbox GL**, implementing interactive maps, routing, dynamic layers, spatial visualization, and analytical dashboards for large-scale data exploration.
- Contributed to enterprise applications deployed on **AWS** using Docker-based development workflows, collaborating across multiple client projects while maintaining scalable, maintainable, and reusable codebases.

Project Work

Karu – Film Workspace

- Architected and developed a full-stack film production workspace enabling screenplay writing, project management, role planning, and production workflows within a unified application.
- Implemented **end-to-end encryption** using the **Web Crypto API** with password-derived cryptographic keys, ensuring screenplay content is encrypted client-side and only ciphertext is stored on backend services.
- Built a **Fountain-compatible screenplay editor** supporting professional screenplay formatting, auto-save, project organization, and encrypted synchronization across devices.
- Designed secure authentication, encrypted storage, and scalable REST APIs for project management, collaboration, and access control while maintaining zero plaintext exposure on the server.
- Engineered a modular architecture that separates presentation, business logic, and encrypted data handling, providing a secure foundation for future real-time collaboration features.

Education

KMG COLLEGE OF ARTS & SCIENCE

B COM - COMPUTER APPLICATION

2020 – 2023

CGPA - 7.4